Automatic Detection of Laghu–Guru Patterns in Sanskrit Chandas

Objective Develop a program that analyzes a Sanskrit shloka written in Devanagari (UTF-8) and identifies the Laghu (L) and Guru (G) syllable pattern. Students must process the input shloka, detect the syllable structure, generate the Laghu–Guru sequence,.

Provided Resources

1. Input Files (Shloka Files)

Dataset Link: https://github.com/nikkkitaaaaaa/sanskrit-chandas-dataset/tree/main/dataset
Files included:
• gayatri_mantra.txt
• totakashtakam.txt
• bhavani_ashtakam.txt
Each file contains the shloka with one pada per line.

2. GIven Shlokas Specifications

Rules File Link: https://github.com/nikkkitaaaaaa/sanskrit-chandas-dataset/tree/main/%20rules

3.Laghu-Guru Classification Rules
https://www.sanskritacademy.org/swaras-practice/
https://santanadharma.fandom.com/wiki/Sanskrit_prosody
Students must implement the Laghu–Guru detection rules from this file.

4.Output Requirement The program must generate an output text file. Example (output.txt): Line 1: LLGLLGLL Line 2: LLGLLGLL...
For each input file an output file should be given for eg. totakashtakam_output.txt...

Deliverables •

Source Code
• Generated output .txt file
• Brief explanation of the algorithm

Github Link: https://github.com/nikkkitaaaaaa/sanskrit-chandas-dataset.git